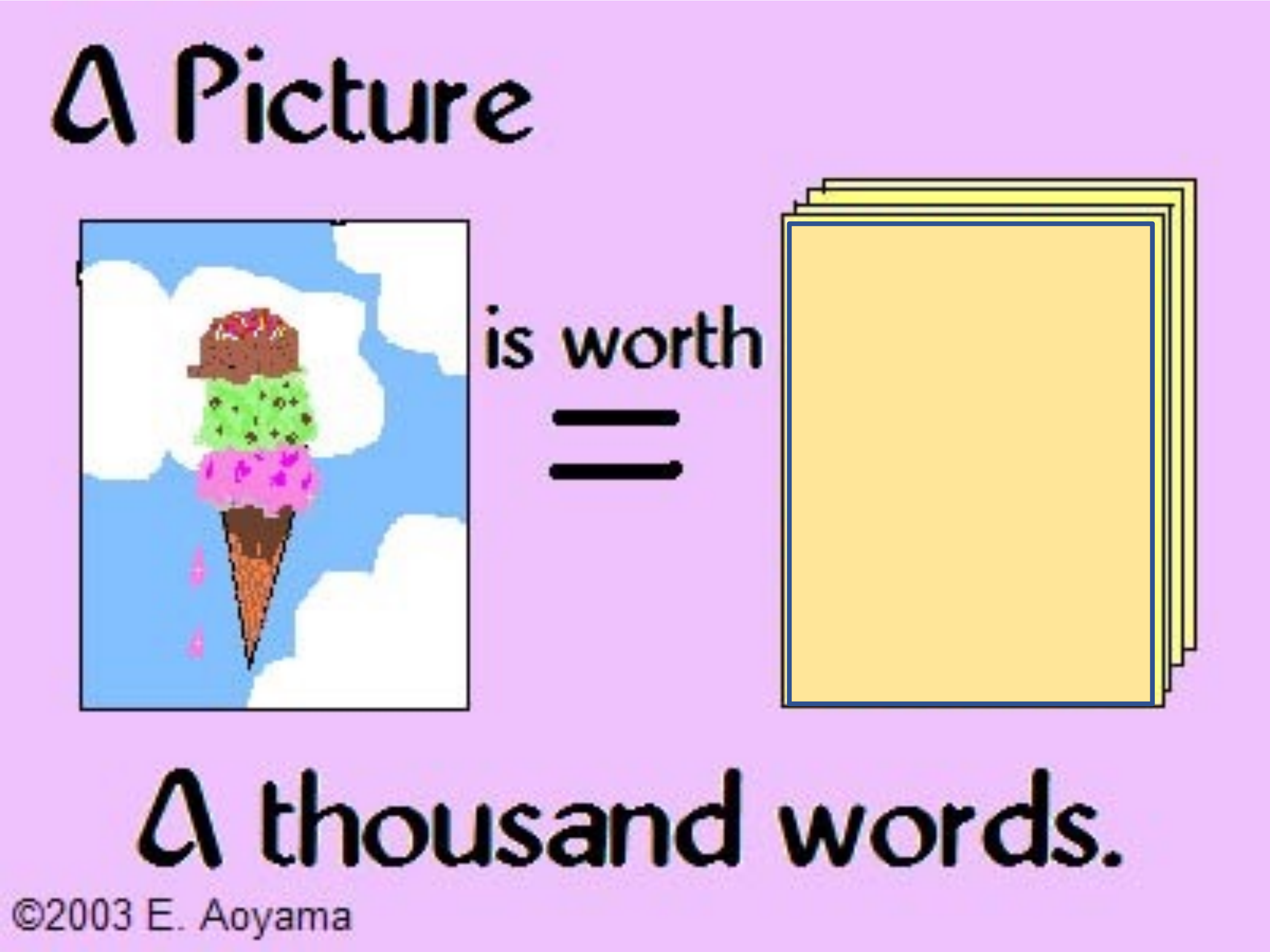
Topic In Computer Vision

Outline

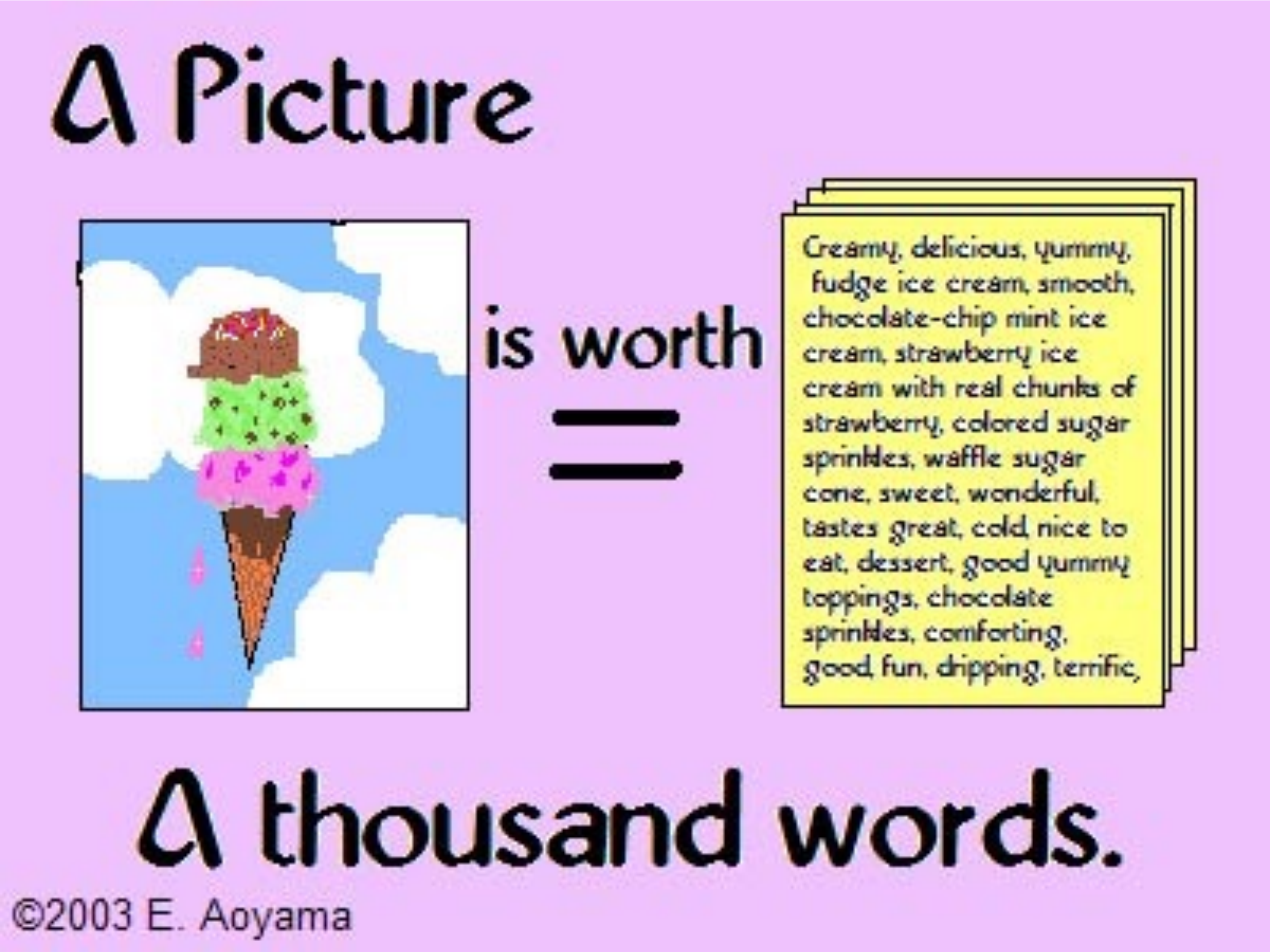
- What is (computer) vision?
- Why computer vision?
- Where are we now?
- What is it related to?

A Picture is Worth 1000 Words



http://oels.byu.edu/student/idioms/proverbs/a_pic_thousand_words.html

A Picture is Worth 1000 Words



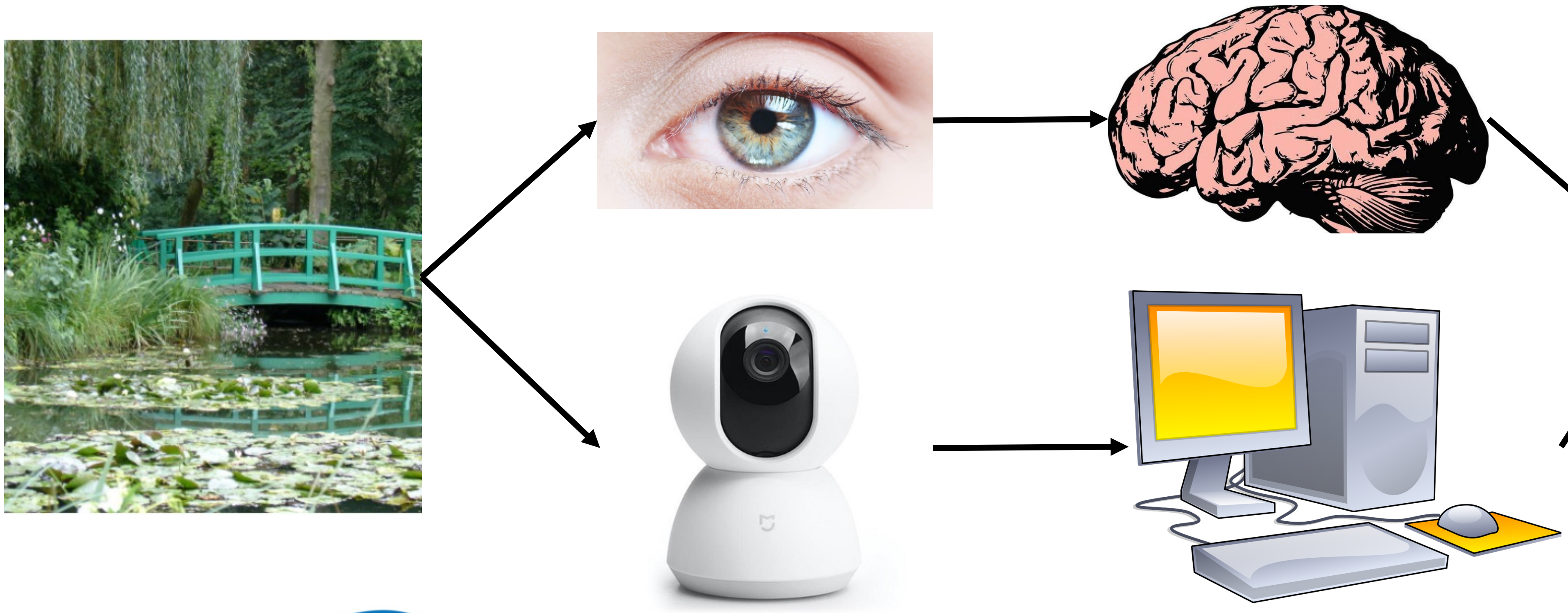
http://oels.byu.edu/student/idioms/proverbs/a_pic_thousand_words.html

What is (computer) vision?

Computer Vision

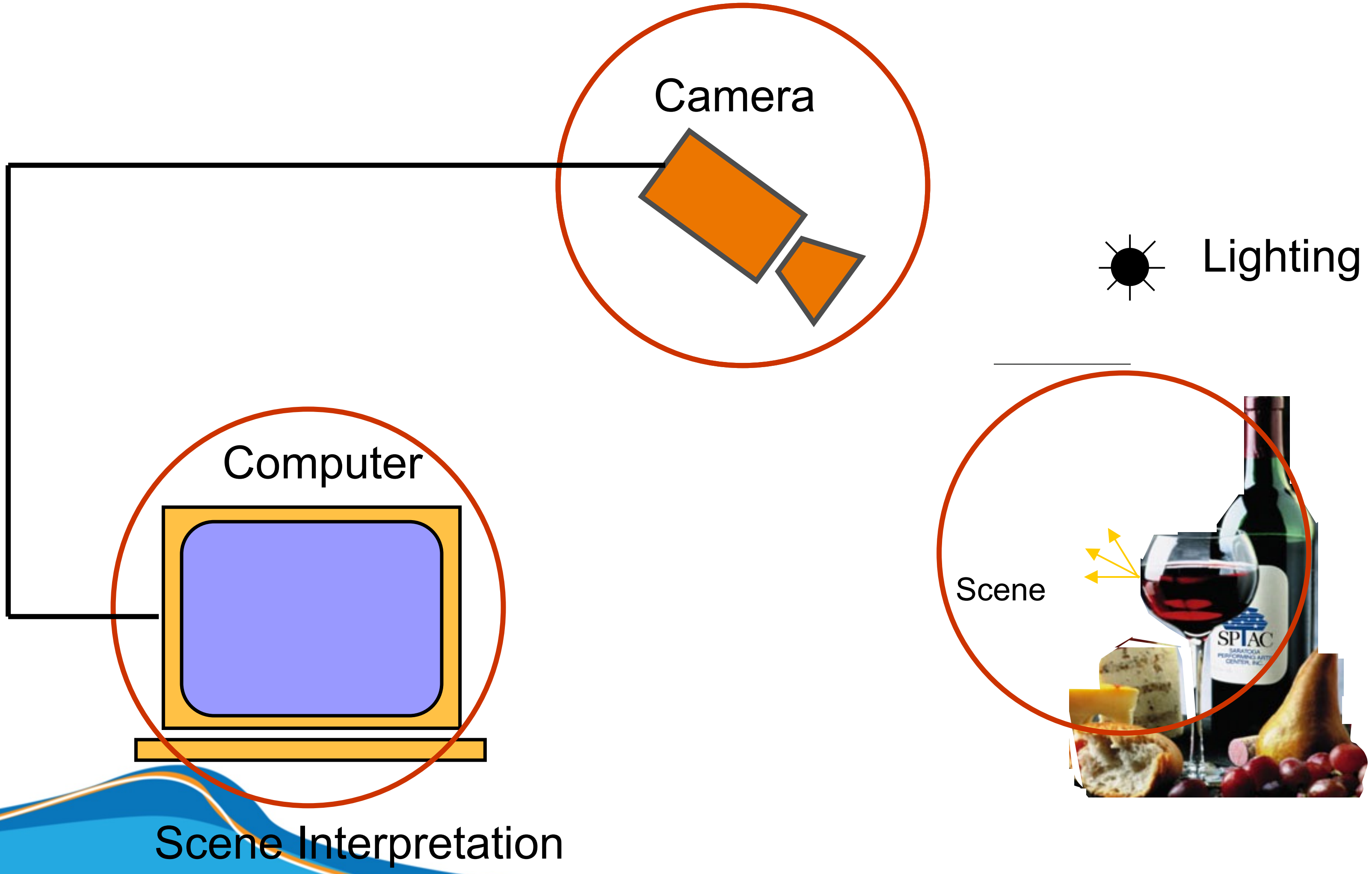
What is (computer) vision?

Image (or video) | Sensing device | Interpreting | Interpretations



garden,
spring,
bridge, water,
trees, flower,
green, etc.

Components of a computer vision system



Srinivasa Narasimhan's slide

Why computer vision?

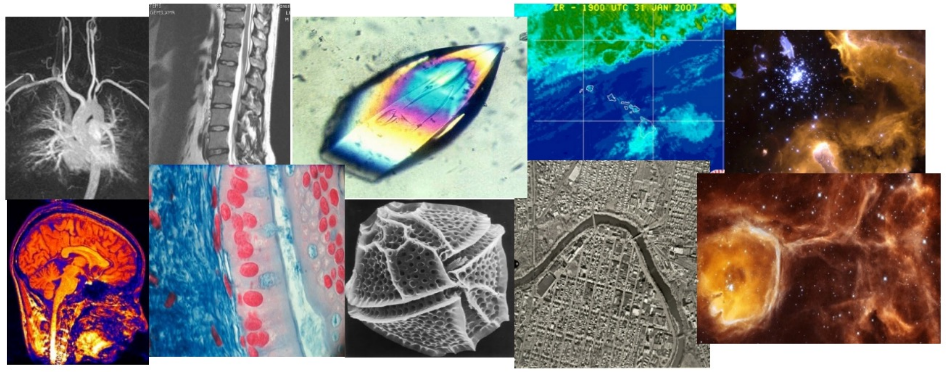
Topics in Computer Vision

Why computer vision?

- Vision is useful: Images and video are everywhere!

Google Image Search · Picasa · flickr · webshots · picsearch · YouTube Broadcast Yourself

Surveillance and security

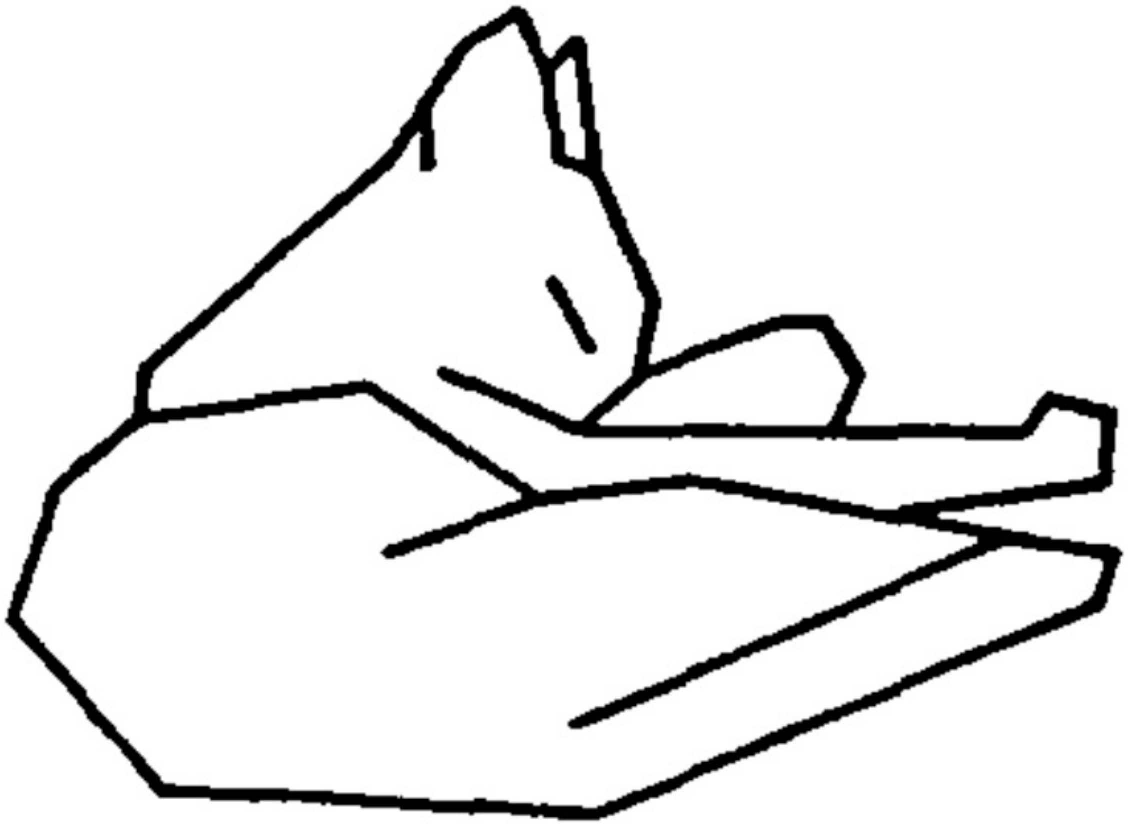


Medical and scientific images

Vision is easy for humans

Vision is easy for humans

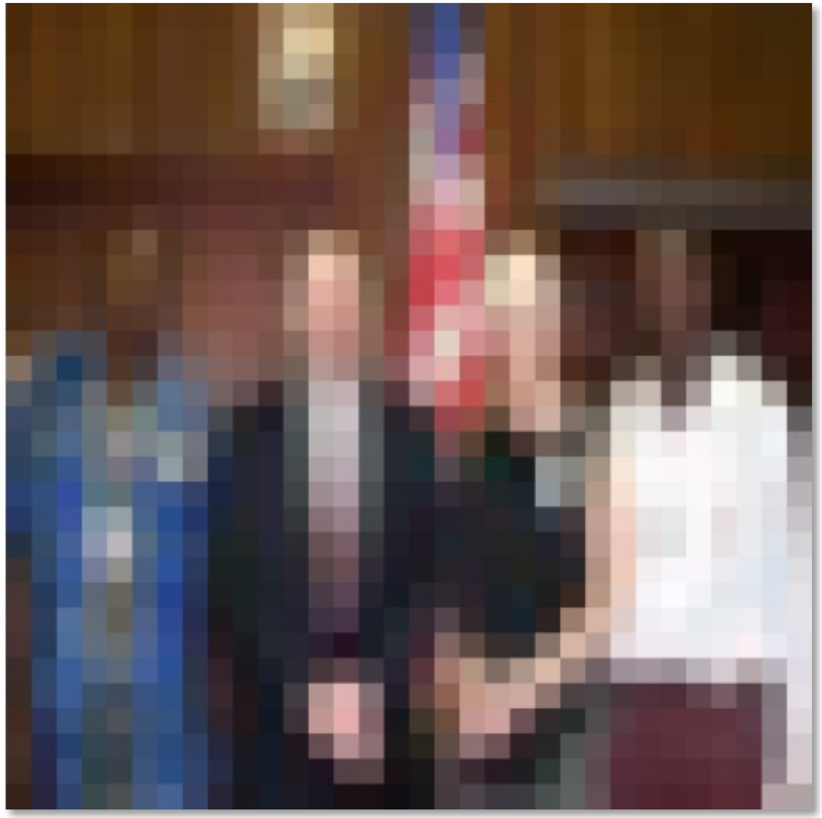
Vision is easy for humans



Attneave's cat

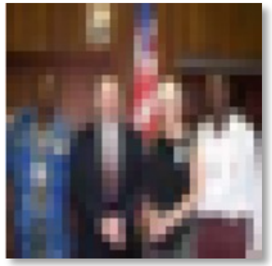
Vision is easy for humans

Vision is easy for humans



Source: “80 million tiny images” by Torralba, et al.

Vision is easy for humans



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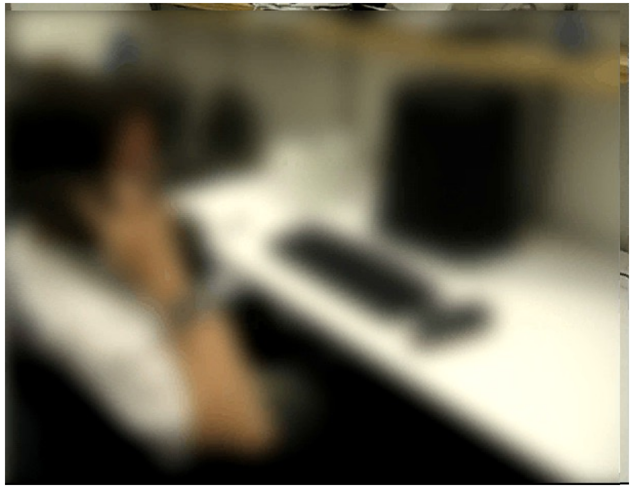


… but not always

[Sinha and Poggio, *Nature*, 1996]

Vision is hard: Images are ambiguous

Vision is hard: Objects blend together

Vision is hard: Objects blend together

Vision is hard: Concepts have variance

The many faces of intra-class variance

Viewpoint variation

Illumination

Scale

The many faces of intra-class variance

Shape variation

Occlusion

Background clutter

The many faces of intra-class variation

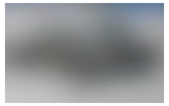
Vision is hard: Concepts are subtle

Tenessee Warbler

Orange Crowned Warbler

https://www.allaboutbirds.org

Vision is hard: local ambiguity



slide credit: Fei-Fei, Fergus & Torralba

Challenges: complexity

- Millions of pixels in an image
- 30,000 human recognizable object categories
- 30+ degrees of freedom in the pose of articulated objects (humans)
- 300+ hours of new video on YouTube per minute
- …
- About half of the cerebral cortex in primates is devoted to processing visual information [Felleman and van Essen 1991]

What the input looks like

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What we see

What a computer sees

The “summer vision project”

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system.

The “summer vision project”

<u>Goals - General</u>

The primary goal of the project is to construct a system of programs which will divide a vidisector picture into regions such as

likely objects

likely background areas

chaos.

We shall call this part of its operation FIGURE-GROUND analysis.

It will be impossible to do this without considerable analysis of shape and surface properties, so FIGURE-GROUND analysis is really insepar-able in practice from the second goal which is REGION DESCRIPTION.

The final goal is OBJECT IDENTIFICATION which will actually name objects by matching them with a vocabulary of known objects.

The big reveal

"… in the 1960s almost no one realized that machine vision was difficult… The common and almost despairing feeling of the early investigators like B.K.P. Horn and T.O. Binford was that practically anything could happen in an image and furthermore that practically everything did."
--- Marr, 1982

Perception is the big problem

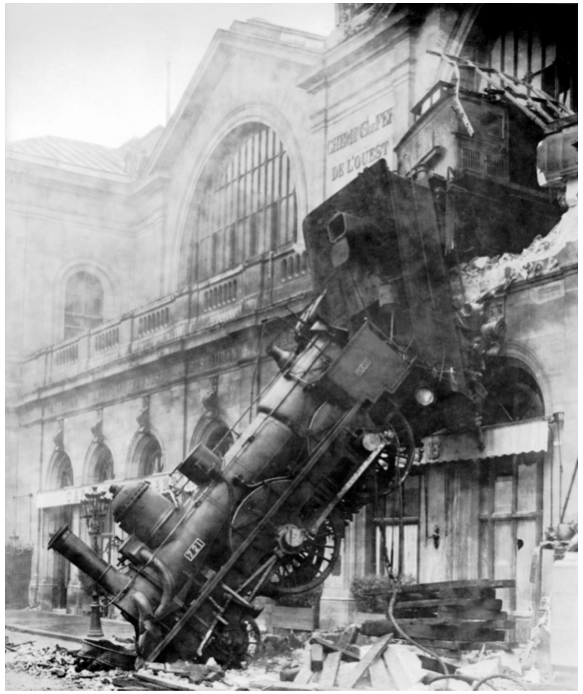
Our first foray into Artificial Intelligence was a program that did a credible job of solving problems in college calculus. Armed with that success, we tackled high school algebra; we found, to our surprise, that it was much harder. Attempts at grade school arithmetic provide problems of current research interest. An exploration of the child’s world of blocks proved insurmountable, except under the most rigidly constrained circumstances.

It finally dawned on us that the overwhelming majority of what we call intelligence is developed by the end of the first year of life.”

– Marvin Minsky, 1977

The goal of computer vision

- To bridge the gap between pixels and "meaning"

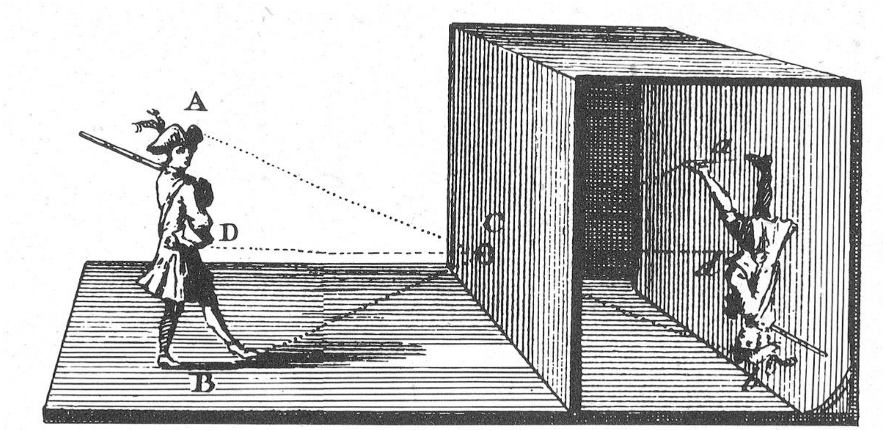


What we see

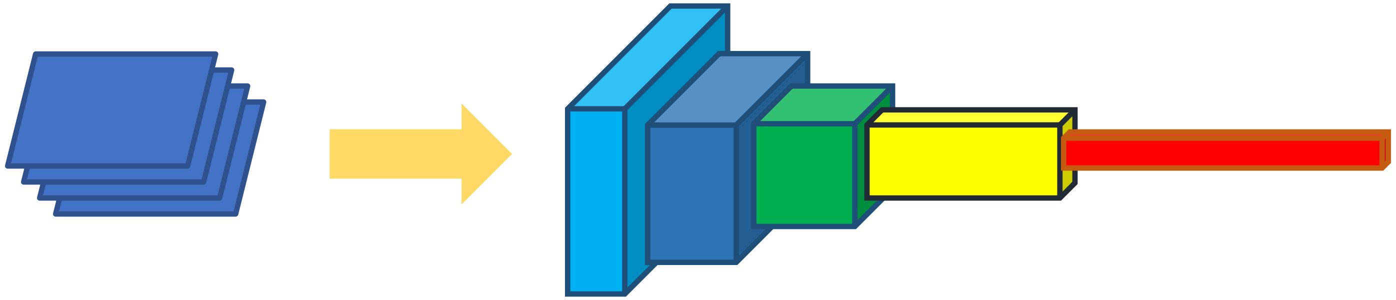
0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

Cues to help us



The physics of image formation



Statistics and machine learning

Where are we now?

Topics in Computer Vision

Deployed: depth cameras

https://realsense.intel.com/stereo/

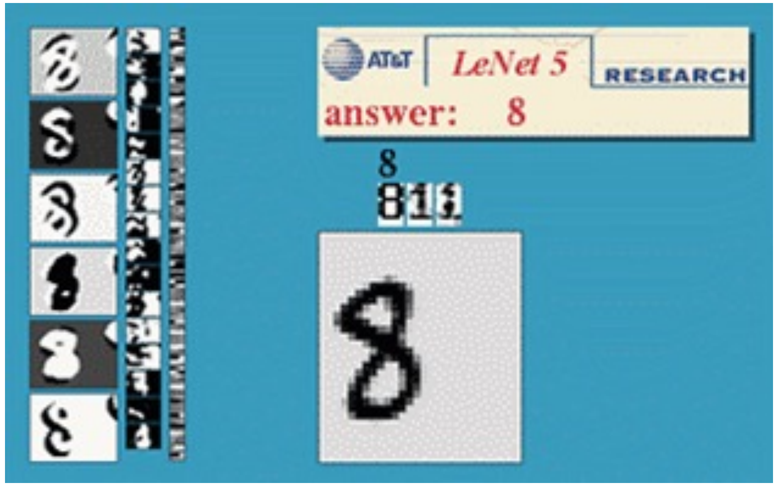
Microsoft Kinect

Deployed: shape capture

The Matrix movies, ESC Entertainment, XYZRGB, NRC

Source: S. Seitz

Deployed: Optical character recognition (OCR)



Digit recognition, AT&T labs
http://www.research.att.com/~yann/

License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

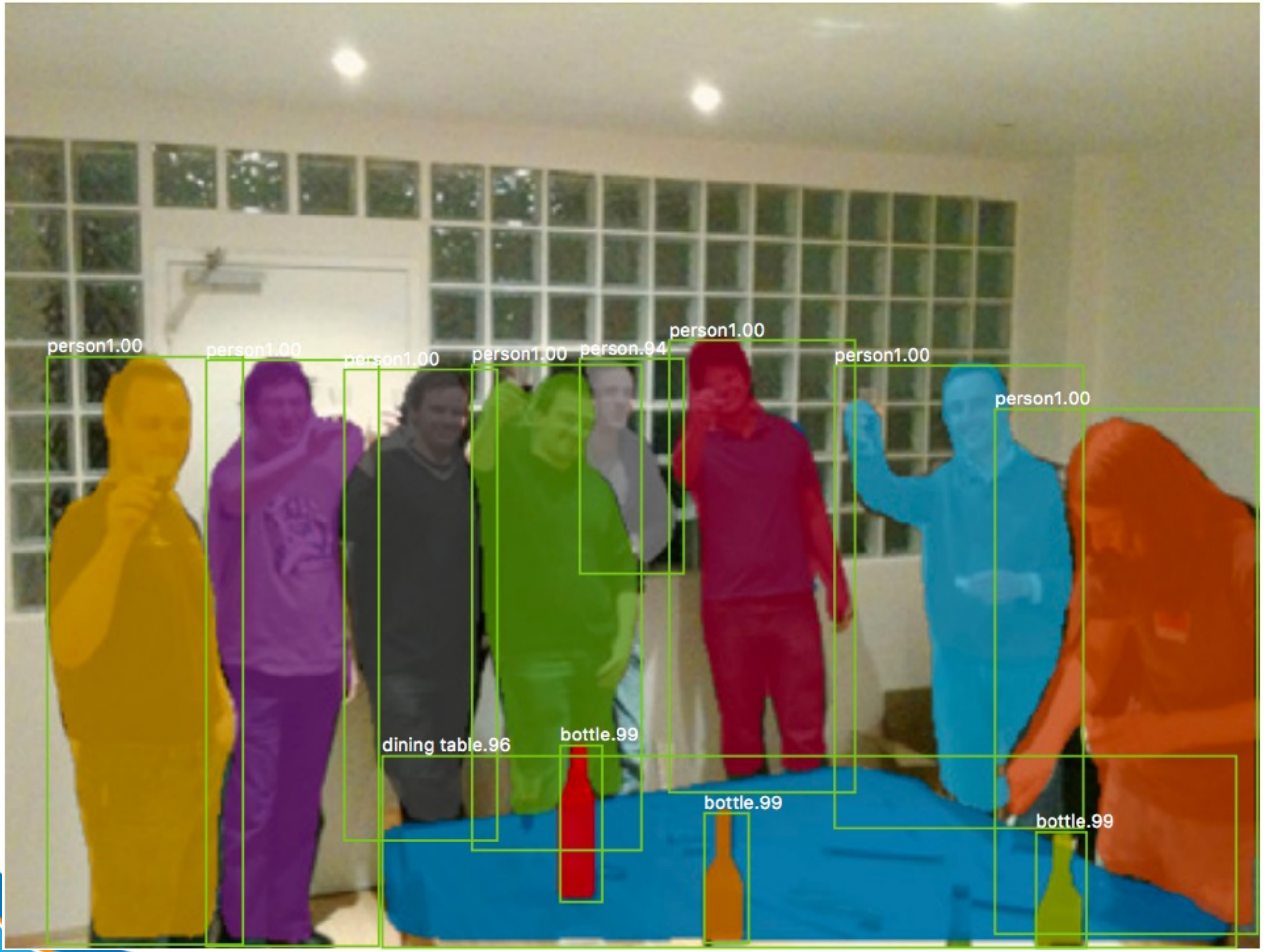
Automatic check processing

Source: S. Seitz

Deployed: Face detection

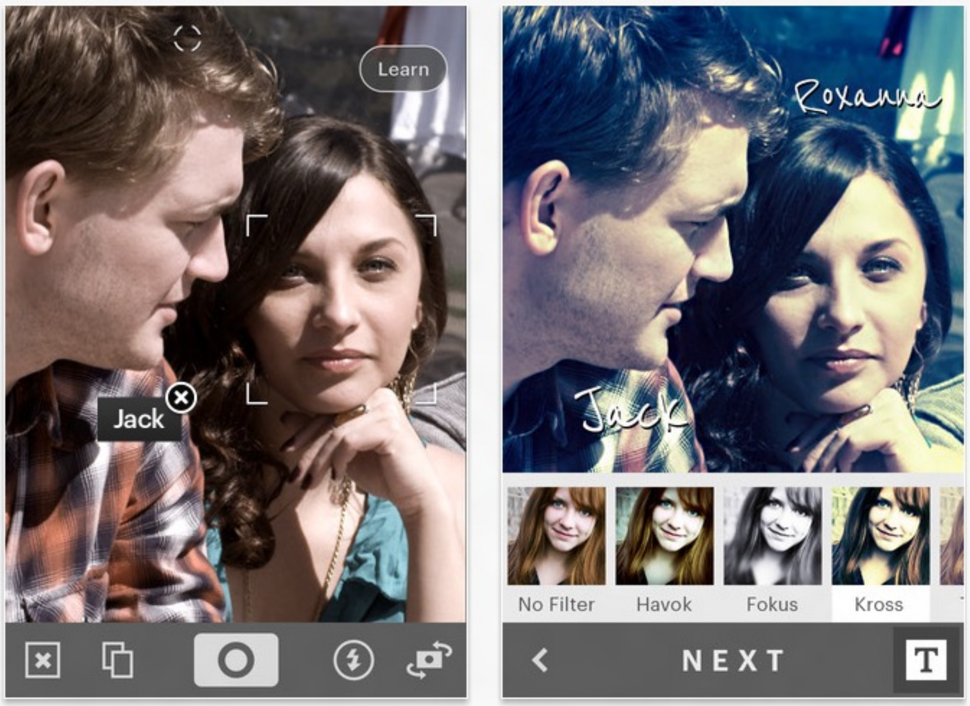
- Many new digital cameras now detect faces
  - Canon, Sony, Fuji, …

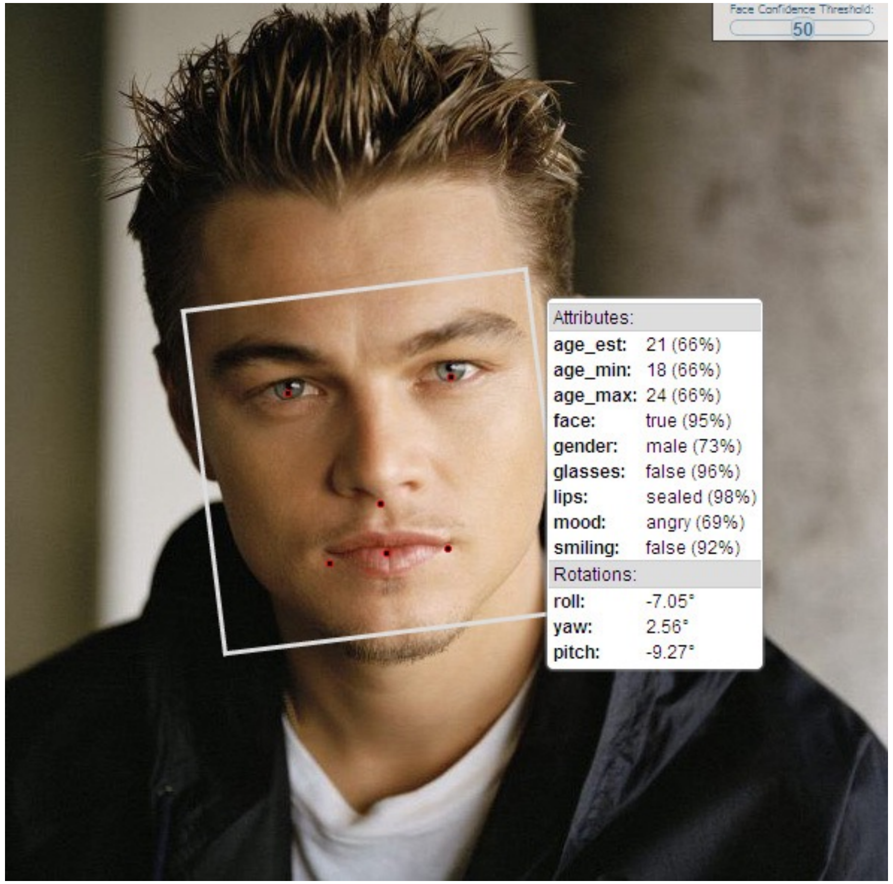
Significant progress: Recognizing objects



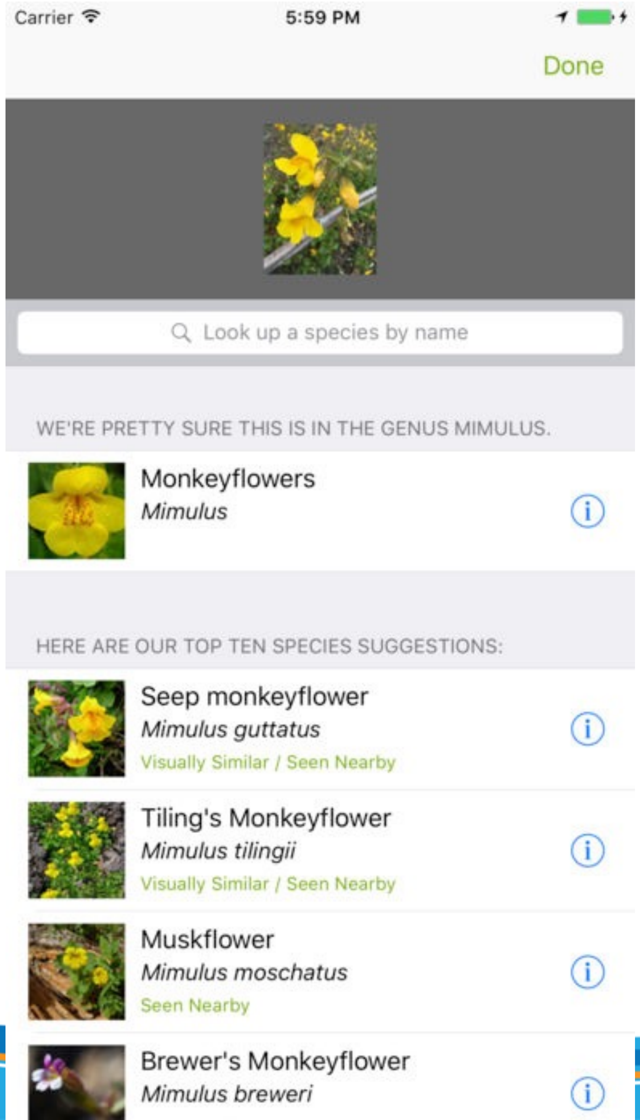
Mask R-CNN. Kaiming He, Georgia Gkioxari, Piotr Dollar, Ross Girshick. ICCV 2017

Significant progress: Face Recognition

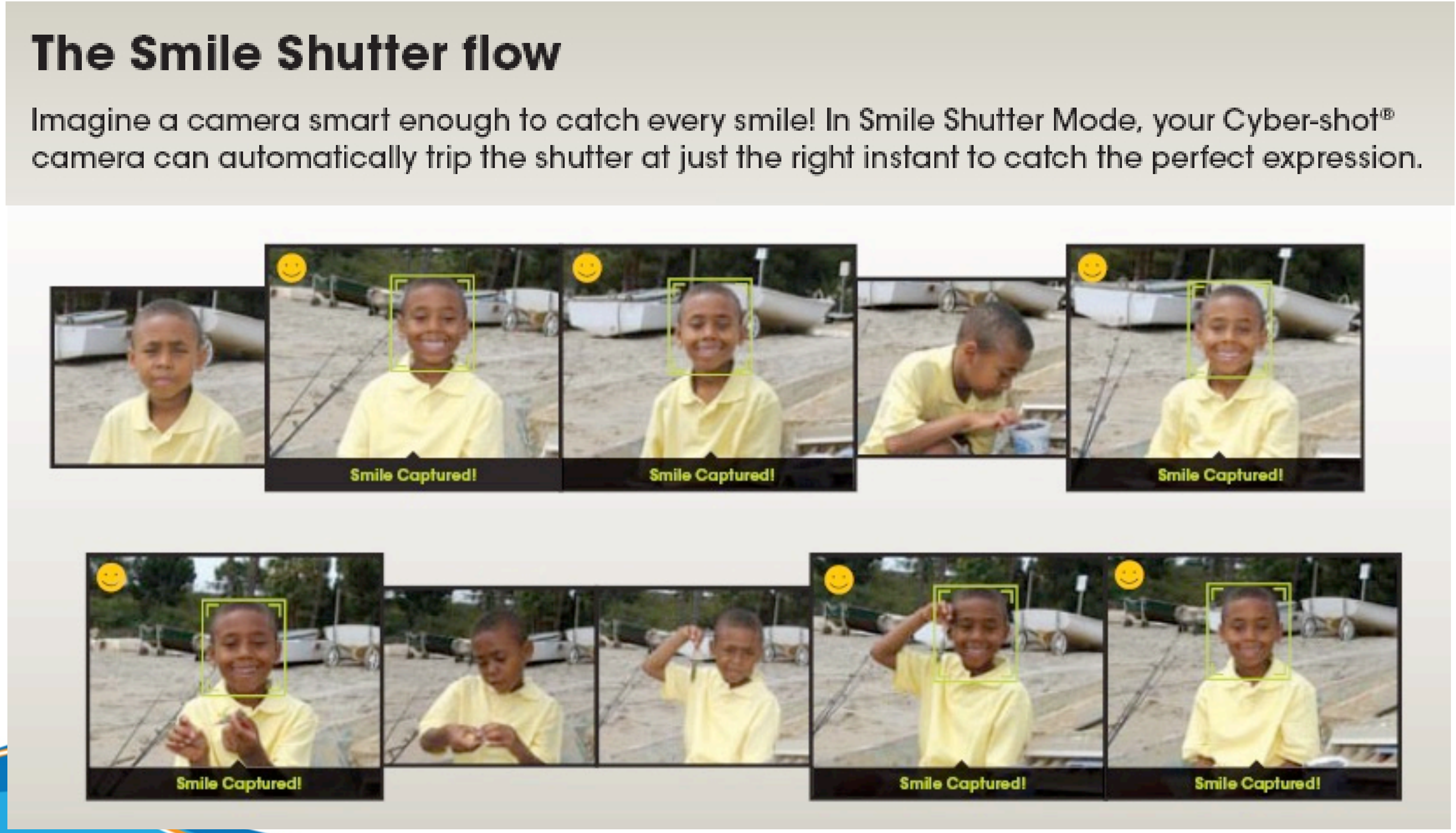




Significant progress: Species recognition

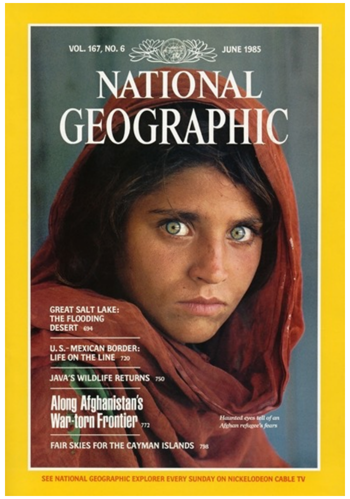


Smile Detection

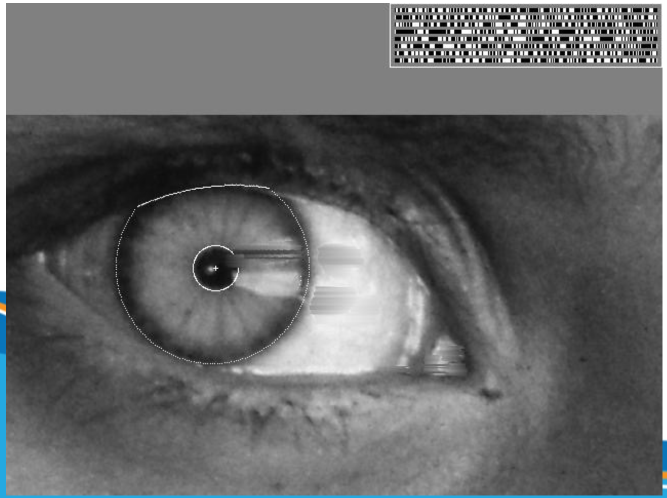
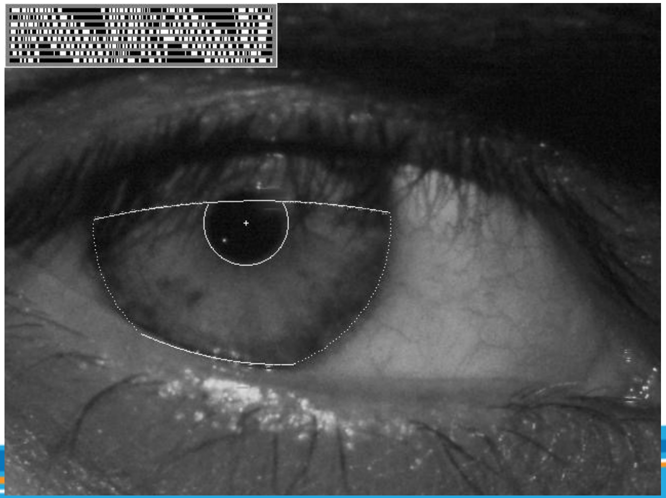


[Sony Cyber-shot® T70 Digital Still Camera](#)

Biometrics



"*How the Afghan Girl was Identified by Her Iris Patterns*" Read the story

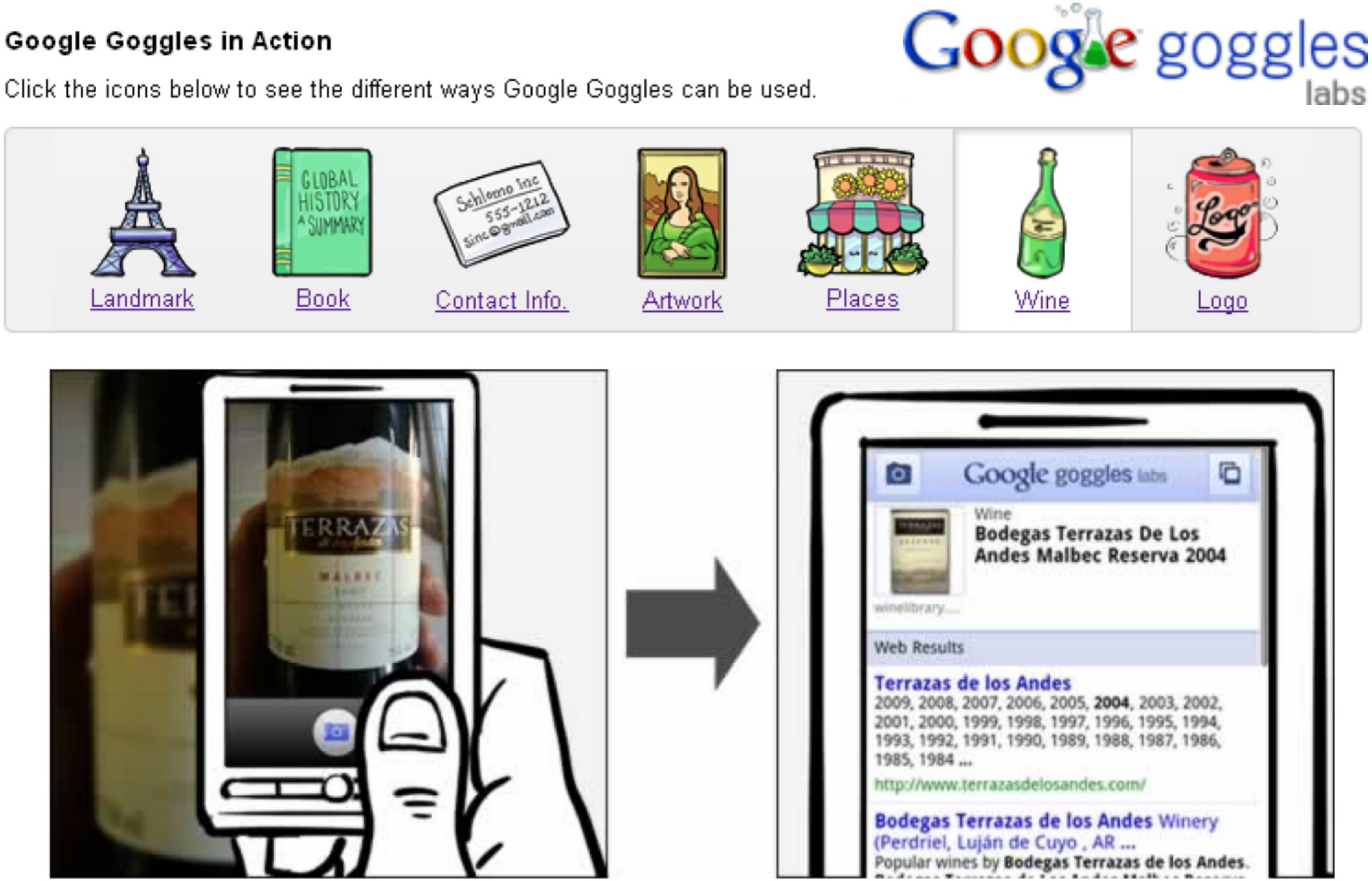



Login without a password…

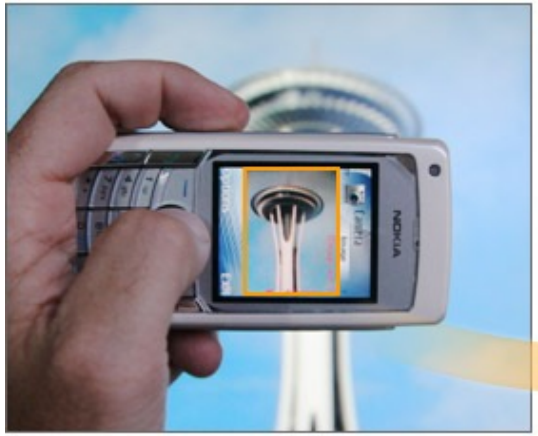
Fingerprint scanners on many new laptops, other devices

Face recognition systems now beginning to appear more widely
http://www.sensiblevision.com/

Mobile visual search: Google Goggles



Object recognition (in mobile phones)



[Point & Find](), [Nokia]()

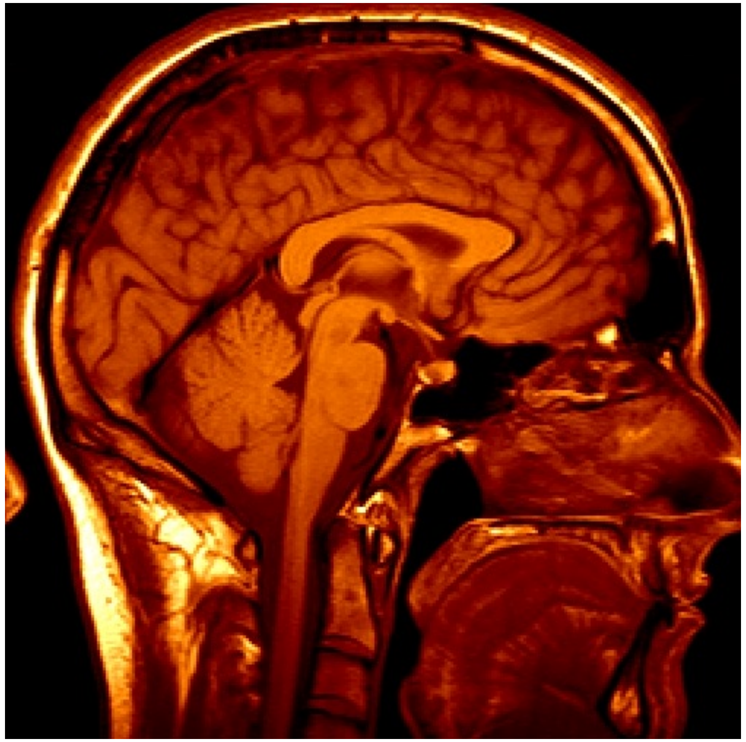
[Google Goggles]()

Object recognition (in supermarkets)

LaneHawk by EvolutionRobotics

"A smart camera is flush-mounted in the checkout lane, continuously watching for items. When an item is detected and recognized, the cashier verifies the quantity of items that were found under the basket, and continues to close the transaction. The item can remain under the basket, and with LaneHawk,you are assured to get paid for it… "

Medical imaging



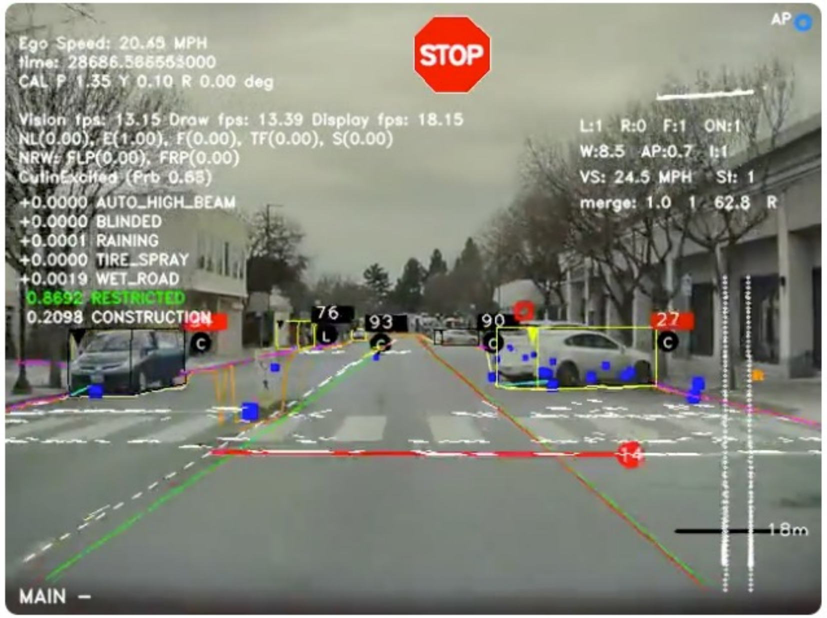
3D imaging
MRI, CT

Image guided surgery
[Grimson et al., MIT]

Automotive safety

- Mobileye
  - Vision systems currently in high-end BMW, GM, Volvo models
  - By 2010: 70% of car manufacturers.
  - Video demo

Self-driving Cars e.g. Google’s Waymo, Tesla …



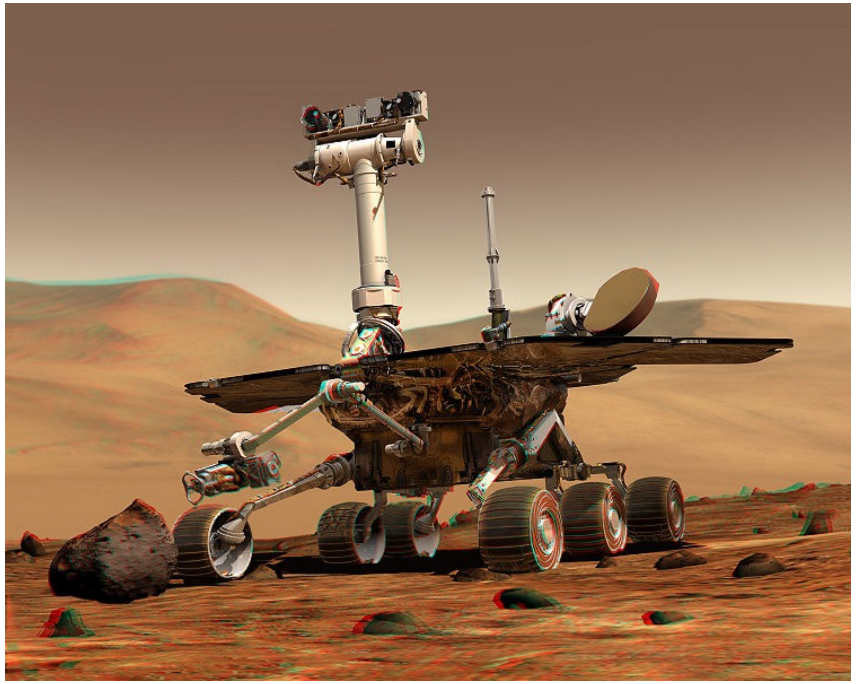
Oct 9, 2010. "Google Cars Drive Themselves, in Traffic". *The New York Times*. John Markoff
June 24, 2011. "Nevada state law paves the way for driverless cars". *Financial Post*. Christine Dobby
Aug 9, 2011, "Human error blamed after Google's driverless car sparks five-vehicle crash". *The Star* (Toronto)

https://www.tesla.com/sites/default/files/images/careers/autopilot/network.mp4

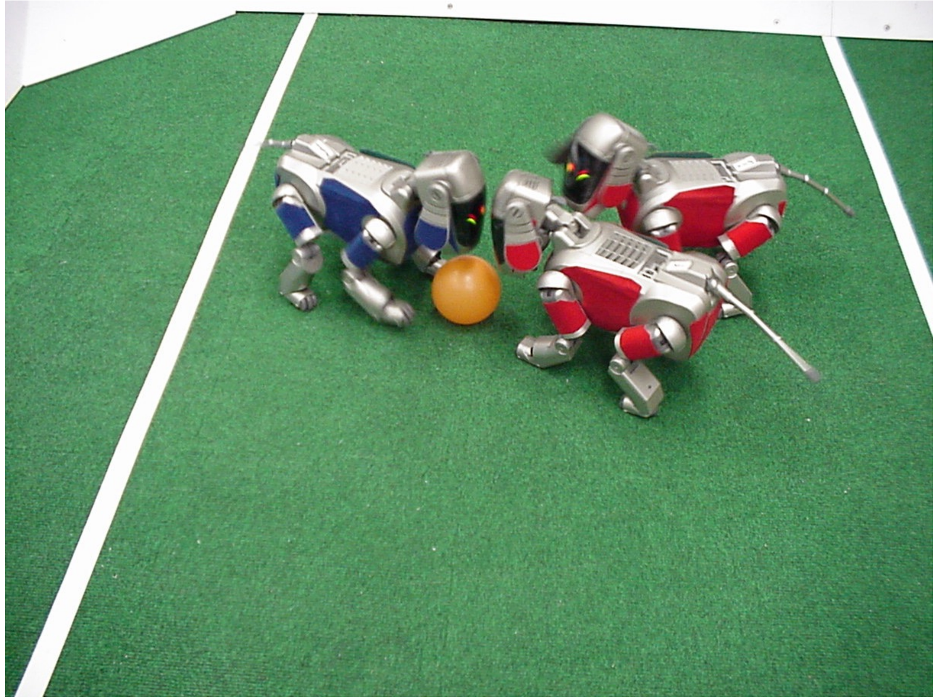
Industrial robots

Vision-guided robots position nut runners on wheels fit@hcmus

Robotics



NASA's Mars Spirit Rover
http://en.wikipedia.org/wiki/Spirit_rover



http://www.robocup.org/

Vision in space

NASA'S Mars Exploration Rover Spirit captured this westward view from atop a low plateau where Spirit spent the closing months of 2007.

Vision systems (JPL) used for several tasks

- Panorama stitching
- 3D terrain modeling
- Obstacle detection, position tracking
- For more, read "Computer Vision on Mars" by Matthies et al.

Vision-based interac.on (and games)

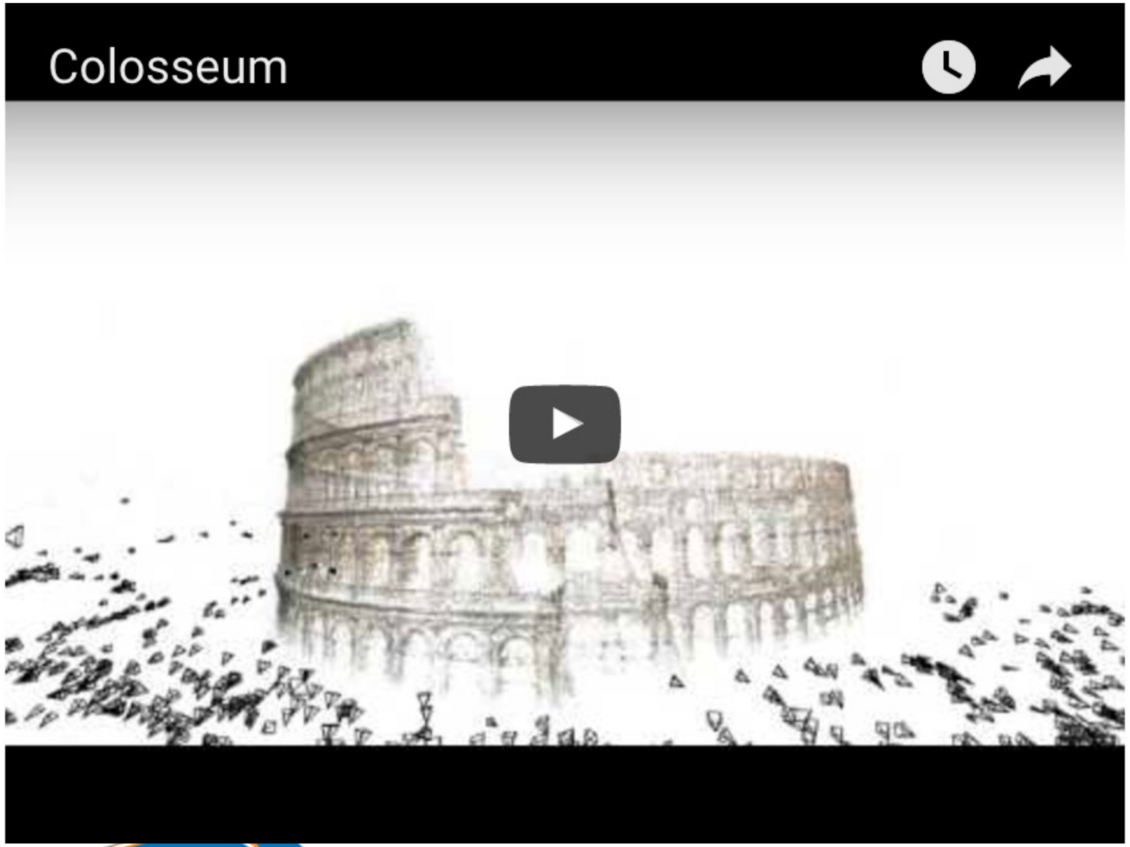
Microsoft's Kinect

Sony EyeToy

Assistive technologies

Source: S. Seitz

Established technology: 3D Models of the world



Building Rome in a Day.
Sameer Agarwal, Noah Snavely, Ian Simon, Steven M. Seitz and Richard Szeliski.
ICCV, 2009, Kyoto, Japan.

Augmented Reality and Virtual Reality

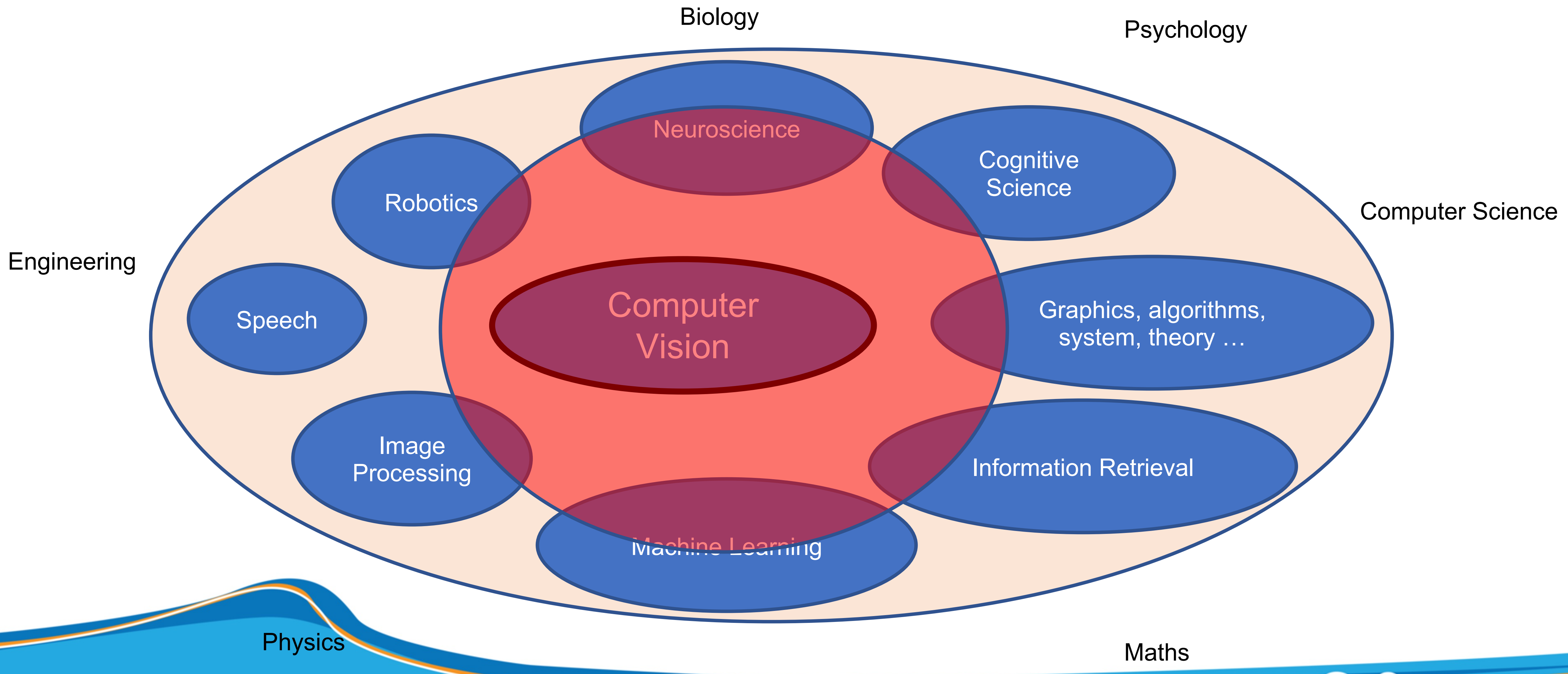
Magic Leap, Oculus, Hololens, etc.

What is it related to?

Topics in Computer Vision

fit@hcmus

What is it related to?



Questions and Answers